

Leo Sun

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Education

Massachusetts Institute of Technology (MIT)

B.S. in Artificial Intelligence with a focus on Robotics

Cambridge, MA

Exp. May 2028

Cumulative GPA: 5.0/5.0

Relevant Courses: Robotics: Science and Systems, Robotic Manipulation, Dynamical System Modeling and Controls, Optimization, Deep Learning, Machine Learning, AI Inference, Algorithms, Probability & Statistics, Linear Algebra, Differential Equations

Involvement: MIT Motorsports, Track and Field (Pole Vault), Phi Delta Theta Fraternity

Experience

Seasats

San Diego, CA

Software Engineer Intern

June 2026 – Present

- Built entire drone swarm software stack for autonomous launch and recovery from USVs, driving PX4 via MAVSDK in Python.
- Developed multi-AprilTag based precision landing with cascaded PID+FF control for **90%** success rate on moving platform.
- Built swarm control with centroid-driven formation setpoints and proximity-based collision avoidance, flying 4 drones at once.
- Created custom Gazebo environment with simulated boat platform to validate entire software stack before flying on hardware.
- Integrated the stack as an async FastAPI service allowing the entire drone swarm to be commanded from the existing Seasats UI.

MIT Motorsports, Driverless

Cambridge, MA

Motion Planning Lead

July 2025 – Present

- Built a C++ midline planner using Delaunay triangulation and cubic-spline interpolation over opposite-color edge midpoints.
- Implemented curvature-based speed profile with PI tracking and OSQP-solved lateral MPC for optimal vehicle path following.
- Created new LiDAR pipeline using Himmelsbach ground removal and 2D Euclidean clustering, improving loop speeds by **40%**.
- Trained a YOLO cone detector and fused it with projected LiDAR centroids in order to produce color-labeled 3D cone positions.
- Built the vehicle state estimation stack using EKF fusing IMU, ground speed sensor, and GPS data into a **100 Hz** pose estimate.

Laboratory for Information and Decision Systems, MIT

Cambridge, MA

Robotics Engineer/Developer

May 2025 – June 2026

- Built a multi-agent pathfinding testbed integrating VMAS simulation and Dockerized ROS 2 stack on DJI RoboMaster S1s.
- Engineered a path-planning algorithm combining Conflict-Based Search with low-level repulsive force collision avoidance.
- Validated algorithm with **400+** random simulations, demonstrating **84%+** reduction in collisions compared to baseline methods.
- Applied an Extended Kalman Filter, fusing multi-camera ArUco detection and robot odometry for multi-robot pose estimation.

Undergraduate Researcher

Jan 2026 – June 2026

- Researched neural combinatorial optimization for CVRP, testing whether a learned policy could improve heuristic-based solvers.
- Applied imitation learning to train an attention network on Lin-Kernighan heuristic moves, beating the baseline **31%** of the time.

Projects

F1TENTH Autonomous Racecar

Cambridge, MA

Software Engineer

Feb. 2026 – May 2026

- Developed ROS 2 autonomy stack on F1TENTH car with Jetson Orin Nano, placing **1st of 10** in a 200m race in MIT's RSS class.
- Implemented RRT-Connect with random shortcutting for **77.5%** faster path generation than A* for navigation with known map.
- Implemented Monte Carlo localization with LiDAR ray-cast sensor model and bicycle-based motion model for **0.09m** RMSE.
- Used color segmentation and homography to locate cones to **0.178m** average error and park in front of cone via pure pursuit.

PoolBot

Cambridge, MA

Motion Planning and Control Engineer

Sep. 2025 – Dec. 2025

- Designed an autonomous pool-playing robot in PyDrake, achieving high-precision shots with avg. angular error of **0.0418** rad.
- Built a shot planner that evaluates **3000** candidate cue-ball angles per turn to generate collision-free, pocket-feasible trajectories.
- Developed the trajectory-generation pipeline, creating multi-keyframe task-space motions and solving IK for consistent striking.
- Co-developed the Drake environment, implementing accurate table geometry, hydroelastic contacts, and friction dynamics.

Skills

Languages: Python, C++, Julia, Java

Libraries/Frameworks: PyTorch, ROS 2, OpenCV, NumPy, SciPy, MAVSDK/MAVLink, PX4, OSQP, FastAPI, Ultralytics YOLO

Tools/Simulation: Gazebo, Drake, VMAS, PyBullet, Unity, Docker, CUDA, NVIDIA Jetson, Git, Linux, RViz, Onshape